

VIThelpdesk

AI-Powered College Helpdesk with Intelligent Tool Selection

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Abstract

VIThelpdesk is an AI-powered college helpdesk assistant developed to automate common student support services. The system identifies user intent using a rule-based approach, selects the appropriate backend tool, and retrieves relevant institutional information using Retrieval-Augmented Generation (RAG) over a structured JSON knowledge base. This approach provides accurate, fast, and cost-effective responses without relying on external API-based knowledge retrieval.

Project Objectives

- Automate college student services and administrative query resolution.
- Detect precise user intent accurately using structured rule matching.
- Execute correct backend tools and fetch context-specific information.
- Deliver accurate, structured responses fast with minimal latency.

System Architecture

The chatbot follows a modular client-server architecture consisting of the following components:

- **Frontend:** HTML, CSS and JavaScript provide an interactive chat interface.
- **Backend:** FastAPI handles request routing and manages chatbot sessions.
- **Intent Detection:** A rule-based engine identifies the user's intent from the query.
- **Tool Layer:** Executes operations such as password reset, ticket creation, ticket listing and FAQ retrieval.
- **Knowledge Base:** Structured JSON documents indexed using FAISS for RAG.
- **Database:** SQLite stores user accounts and support ticket information.

Approach Comparison

To optimize reliability and response consistency, various design architectures were evaluated:

- **Rule-Based Approach:** Fast, runs completely locally, and incurs zero cost. (Selected for reliability)
- **LLM-Based Approach:** Highly semantic, but introduces latency and expensive API dependency.
- **Hybrid Approach:** Balanced capability, but suffers from higher architectural complexity and state management overhead.

Evaluation Results

The chatbot was evaluated using 53 test cases covering clear requests, ambiguous queries, multi-step conversations and sensitive inputs.

Metric	Result
Total Test Cases	53
Intent Detection Accuracy	73.6%
Tool Selection Accuracy	96.0%
Response Quality	100.0%

Table 1: Evaluation Summary

```
(venv) sravanthi-murukuri@sravanthi-murukuri-HP-Laptop-15s-fr4xxx:~/ai-support-chatbot$ /home/sravanthi-murukuri/ai-support-chatbot/venv/bin/python /home/sravanthi-murukuri/ai-support-chatbot/test_eval.py
[✓] FAISS Vector Database loaded into memory successfully!
=====
🚀 RUNNING UPGRADED CHATBOT EVALUATION
=====
📊 EVALUATION RESULTS SUMMARY
=====
• Total Test Cases Evaluated      : 53
• Intent Detection Accuracy       : 73.6% (39/53)
• Tool Selection Accuracy         : 96.0%
• Response Quality / Success Rate: 100.0% (53/53)
=====
[✓] Upgraded evaluation completed successfully!
%(venv) sravanthi-murukuri@sravanthi-murukuri-HP-Laptop-15s-fr4xxx:~/ai-support-chatbot$
```

Figure 1: Terminal Output of Evaluation Results

Challenges Faced

- Designing reliable rule-based intent detection for different user phrasings.
- Organizing institutional knowledge into structured JSON for accurate RAG retrieval.
- Preventing unrelated information from being retrieved during semantic search.
- Maintaining conversational state across multi-step interactions.

Future Improvements

- Develop an admin dashboard for managing FAQs and support tickets.
- Add voice-based interaction for accessibility.
- Introduce multilingual support for regional languages.
- Integrate live college notifications and event updates.
- Improve semantic retrieval using a hybrid retrieval strategy.

Conclusion

VIThelpdesk demonstrates an efficient AI-powered college support assistant capable of understanding user intent, selecting appropriate tools, and retrieving relevant institutional information through structured JSON-based RAG. The system achieves fast response times, reliable tool execution, and accurate information retrieval while remaining lightweight, scalable, and independent of external API-based knowledge sources.